

INDIAN MATRIVERSE CONSTRUCTION

A PROJECT REPORT FOR DESIGN THINKING AND METHODOLOGY

Submitted by

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BONAFIDE CERTIFICATE

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INTERNAL EXAMINER- I

INTERNAL EXAMINER-II

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ABSTRACT

In today's fast-paced construction industry, the need for efficient and streamlined communication, coordination, and delivery processes is paramount. Construction projects require seamless coordination between various stakeholders, timely material deliveries, and effective management of resources. To address these challenges, this study proposes the development of a comprehensive online platform tailored specifically for the construction field delivery process.

The envisioned platform leverages cutting-edge technology to create a user-friendly interface that connects suppliers, contractors, and construction site managers in real-time. By integrating features such as real-time inventory tracking, automated order processing, and predictive analytics, the platform enhances the efficiency of material procurement and delivery. Additionally, the system incorporates geolocation services, enabling accurate tracking of deliveries and optimizing route planning for suppliers.

Key functionalities of the proposed platform include:

Supplier Management: Suppliers can register, update their product catalog, and monitor demand fluctuations in real-time. The system assists suppliers in managing inventory levels and automates order fulfillment processes.

Order Processing: Construction site managers can place orders directly through the platform, specifying quantities, delivery schedules, and site locations. The platform automates order processing, ensuring accuracy and eliminating delays caused by manual errors.

Real-time Tracking: Utilizing GPS technology, the platform provides real-time tracking of delivery vehicles. Site managers can monitor the progress of deliveries, receive accurate ETA notifications, and plan construction activities accordingly.

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EXPLOREPHASE

1.1 STEEP ANALYSIS

SOCIAL & DEMOGRAPHICS

Urbanization Trends
Internet Penetration
Digital Literacy
Construction Workforce

TECHNOLOGY

Iot (internet of things)
Cloud computing
Mobile app development
Machine learning

ECONOMY

Market economy
Emerging economy
Developing economy

ENVIRONMENT & NATURE

Sustainable development source
Carbon footprint reduction
Water conservation

POLITICS & LEGAL

Regulatory Compliance
Licensing and Certification
Intellectual Property Rights

**YOUR
DESIGN
CHALLENGE
ISSUE**

Natural disasters such as hurricanes, earthquakes, or floods can severely disrupt the supply chain and damage construction materials	High Impact Frequent fluctuations in material prices due to market demand, geopolitical tensions, or supply shortages can significantly impact project costs.
Uncertain to Occur Small delays in material deliveries that can be easily accommodated within the project timeline.	Likely to Occur Temporary disruptions due to routine maintenance work on roads or infrastructure, causing minor delays in material deliveries Low Impact

1.2 STEEP ANALYSIS PRIORITIZATION

Discuss 2 key trends from the High Impact – Likely to occur quadrant:

1. Significant social trends, like a shift towards sustainable construction practices, can have a high impact. Emphasizing eco-friendly materials might attract a large segment of environmentally conscious buyers.
2. Rapid technological advancements, especially in mobile technology and data analytics, have a high impact. A user-friendly mobile app and sophisticated data analytics can have significantly enhance user experience and engagement.

Discuss 2 trends from the High Impact – Uncertain to occur quadrant

1. Mild fluctuations in economic indicators might have a limited impact. Regular monitoring is necessary, but minor changes may not drastically affect the platform.
2. Stable political conditions in supplier regions have a low impact. However, it's essential to be aware of potential changes that might impact the supply chain.

1.3 STEEP ANALYSIS MATRIX

	URGENT	LESS URGENT
IMPORTANT	Significant social trends like the shift towards sustainable construction practices have high urgency. Emphasizing eco-friendly materials and suppliers should be a priority due to the current market demand. Rapid technological advancements, especially in mobile technology and data analytics, have high urgency. Developing a user-friendly mobile app and implementing advanced data analytics should be prioritized to enhance user experience.	Slow adoption of advanced technologies within the construction industry has low urgency but should be continuously monitored for future integration. Minor demographic differences (age, occupation) have low urgency. While important for market analysis, immediate action might not be necessary. Mild fluctuations in economic indicators have low urgency. Continuous monitoring is necessary, but immediate actions might not be critical.
LESS IMPORTANT	<i>Immediate Attention Needed:</i> A sudden change in user behavior patterns or preferences, even if historically less significant, might demand urgent website adjustments to cater to new user needs promptly. Addressing unexpected technical issues or security vulnerabilities, even if they don't significantly impact the core functionality, is urgent to ensure the platform's stability and user trust.	<i>Monitor for Future Consideration:</i> Minor demographic changes or less impactful social trends can be monitored for future adjustments in marketing strategies. While they might not require immediate changes, being aware of these shifts is essential for long-term planning. Less critical technological advancements can be periodically reviewed. Although they don't demand immediate action, staying updated ensures the platform doesn't fall behind.

1.4 SYNTHESIS: MAKING SENSE OF STEEP ANALYSIS AND STRATEGIC PRIORITIES

ASSESSMENT QUESTIONS	SYNTHESIS: SENSE MAKING
What relationships among the trends do you perceive? How are they related? Why are these relationships important	Technology and User Experience: Advancements in technology (like mobile apps, AI, and data analytics) directly impact user experience. A user-friendly mobile app and personalized experiences through data analytics enhance customer satisfaction.
What opportunities and/or challenges need immediate attention going forward for your design challenge? And why?	Market Expansion: Expanding the platform to untapped markets or regions with growing construction activities presents a significant opportunity. It allows the platform to reach new customers and suppliers, increasing revenue streams.
What would it take to create positive change on this issue relating to your design challenge?	Networking and Partnerships: Collaborate with construction industry associations, sustainable material suppliers, and environmental organizations. Partnerships can provide valuable insights and support for eco-friendly initiatives.
Who else would be interested in this issue? Why should they care? What conversations would you have with them?	Conversation Topics: Discuss the benefits of sustainable materials, cost-effectiveness in the long run, and case studies demonstrating successful projects with eco-friendly materials. Provide resources and expert insights on integrating these materials into construction projects.

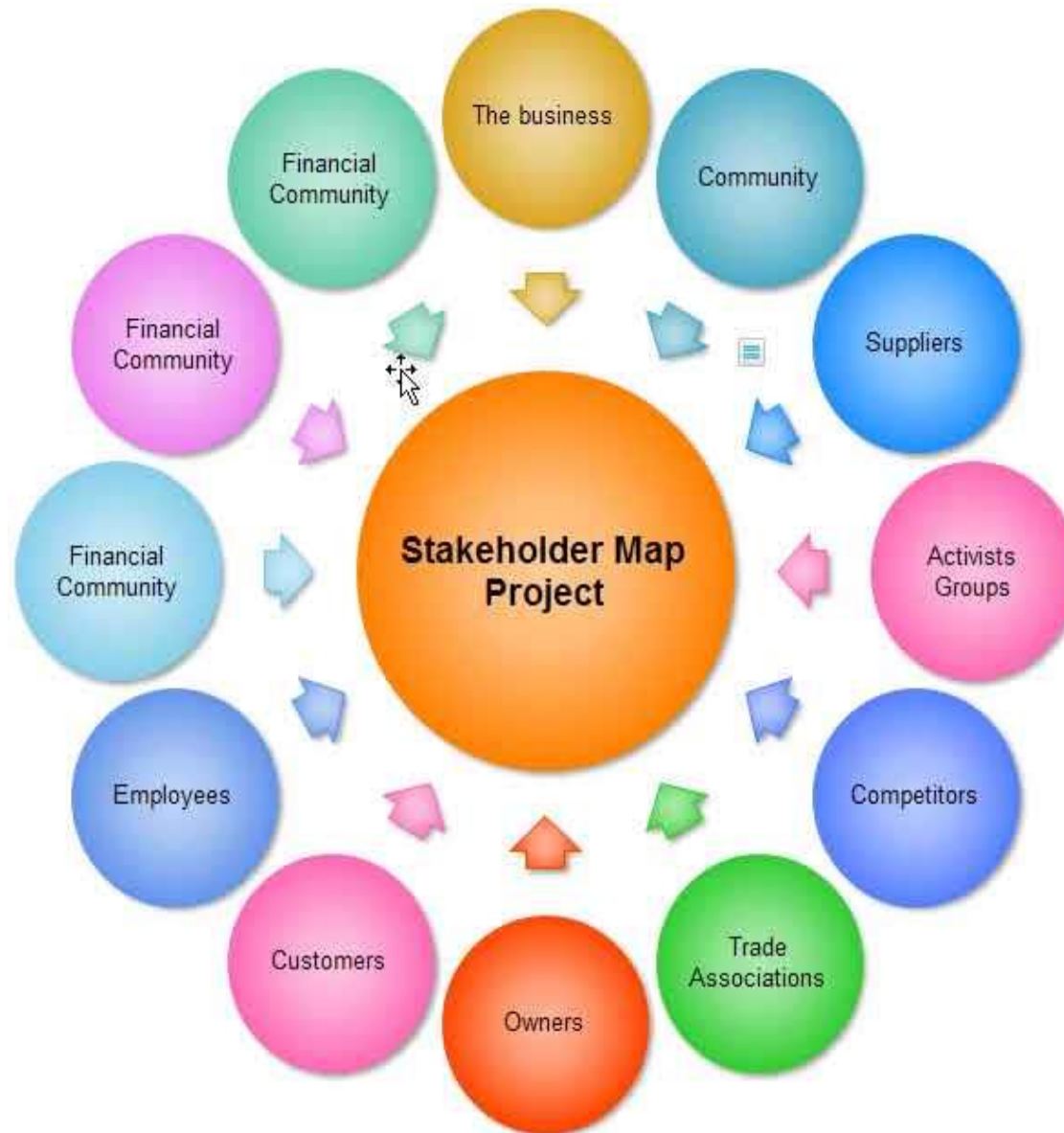
**OBJECTIVES OR
MISSION STATEMENT**

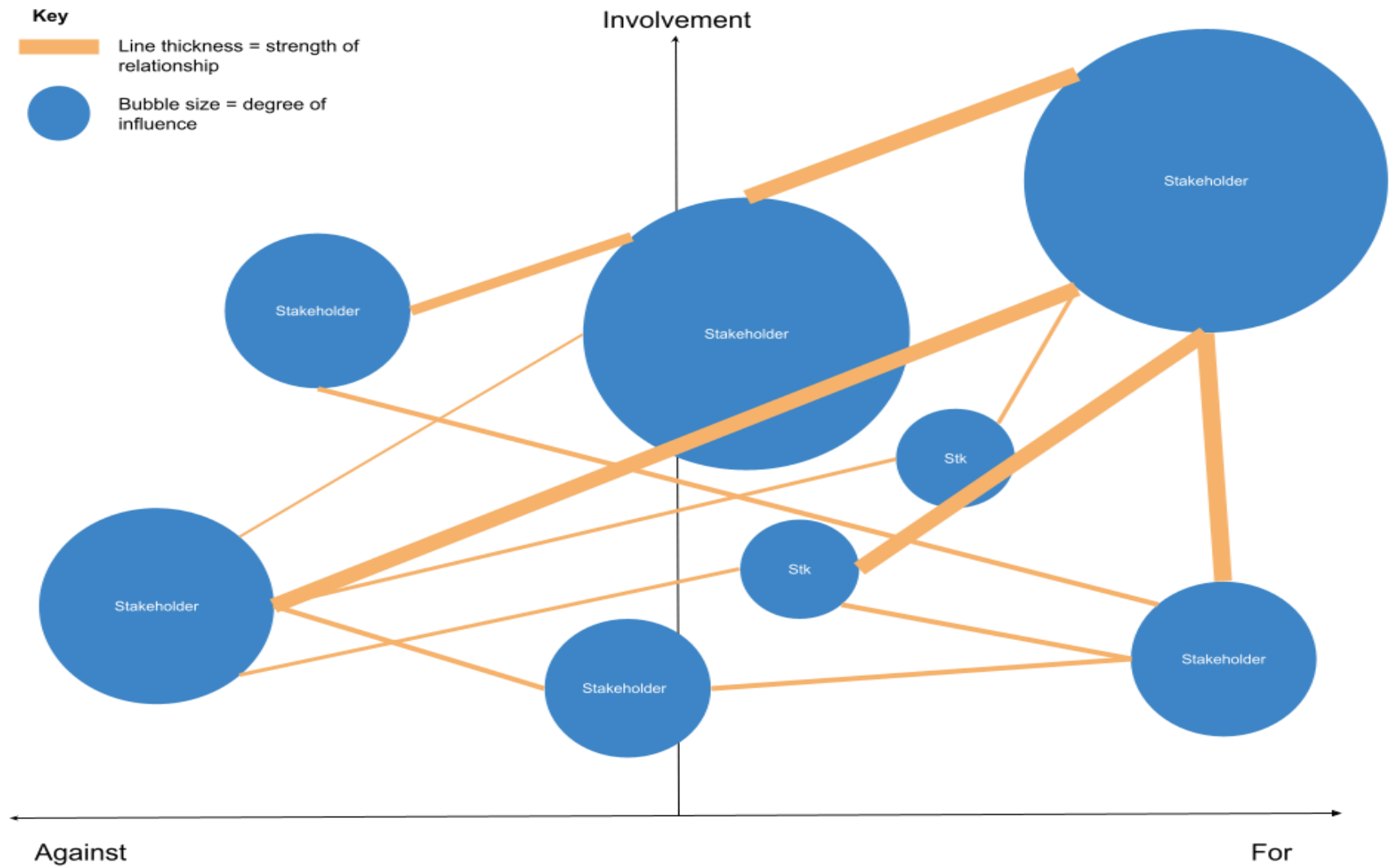
SUPPLIER ENGAGEMENT AND CERTIFICATION:	USER EDUCATION AND AWARENESS:	INCENTIVES FOR SUSTAINABLE CHOICES:	COLLABORATIVE PARTNERSHIPS AND ADVOCACY:
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Strategy: Establish a certification program for suppliers offering sustainable construction materials. Encourage suppliers to obtain recognized environmental certifications (such as FSC, LEED, or ISO) for their products. Implementation: Offer incentives, such as increased visibility on the platform, to certified suppliers. Regularly audit and verify supplier certifications to maintain quality standards	Strategy: Develop educational resources and content to raise awareness among construction professionals about the environmental impact of construction materials. Provide guides, articles, and webinars on sustainable building practices. Implementation: Integrate educational content directly into the platform. Implement email newsletters, webinars, and social media campaigns to disseminate	Strategy: Offer incentives or discounts for users who choose sustainable materials. Implement a rewards program where users earn points for selecting eco-friendly options, which can be redeemed for future purchases. Implementation: Integrate the rewards program into the platform's user accounts. Highlight sustainable products with special badges or labels to	Strategy: Forge partnerships with environmental organizations, government agencies, and industry associations. Collaborate on joint initiatives, awareness campaigns, and policy advocacy to promote sustainable construction practices. Implementation: Organize joint events, workshops, and seminars. Engage in policy discussions, advocating for
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Benefits: Informed users are more likely to choose sustainable materials. Education empowers users to make environmentally conscious decisions, promoting the use of eco-friendly materials in construction projects.	information. Ensures that the materials listed on the platform meet recognized sustainability criteria. Provides users with a reliable selection of eco-friendly options, fostering trust and encouraging the use of sustainable materials.	indicate discounts or additional benefits Encourages users to prioritize sustainable choices. Incentives drive user engagement and create a positive feedback loop, promoting the continued use of eco-friendly materials.	regulations that promote the use of sustainable materials in construction projects. Strengthens the platform's credibility and influence. Collaborative efforts amplify the message of sustainability, reaching a wider audience. Advocacy initiatives can drive policy changes that support eco-friendly practices industry-wide.
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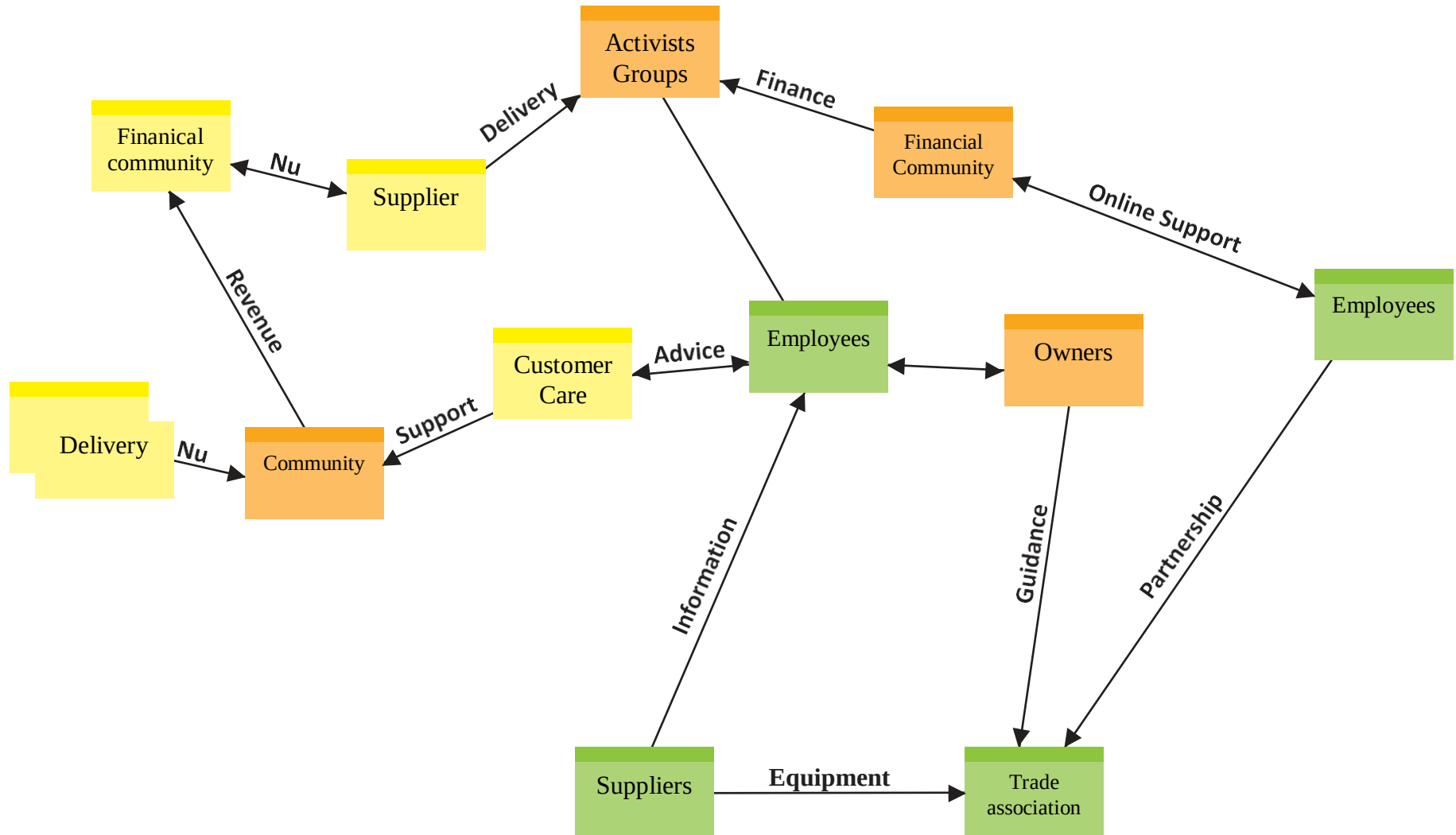
1.5 STAKEHOLDER MAPPING MATRIX





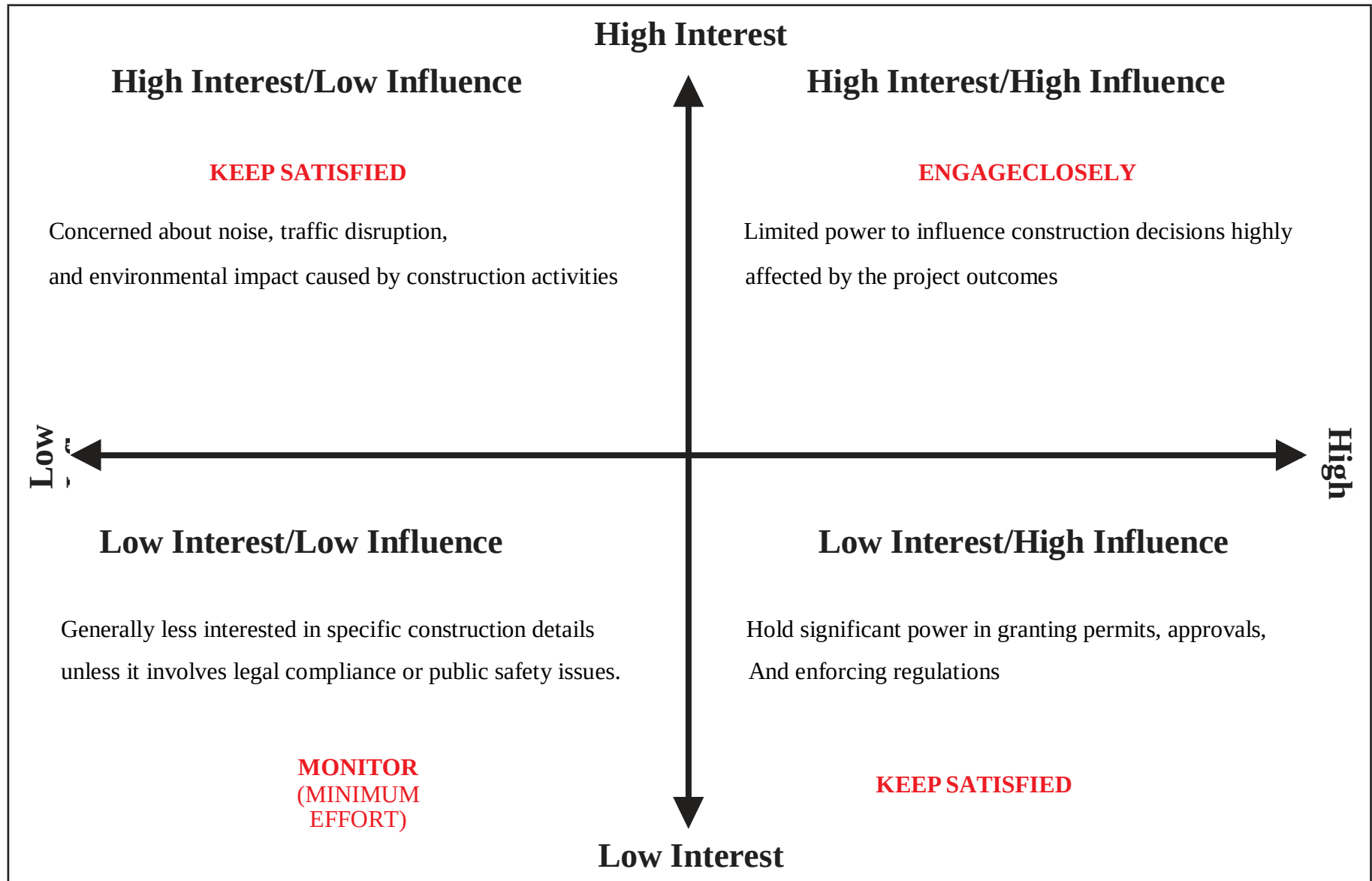
1.6 STAKEHOLDER LINKS & RELATIONSHIP MAPPING TEMPLATE

(Define the links and relationships between stakeholders)



1.7 STAKEHOLDER PRIORITY MAPPING MATRIX

(Develop engagement Strategies)



1.8 STAKEHOLDER ANALYSIS & ENGAGEMENT STRATEGY

Key Stakeholders	Relationships	Stakeholder's Interest(s) in the Design Challenge	Impact Assessment	Strategies to Gain Support or Reduce Obstacles
Partners	Supplier-Construction Professional Relationship	Construction Professionals	Environmental Impact	Stakeholder Engagement
Influencers	Supplier-Manufacturer Relationship:	Suppliers and Manufacturers	Economic Impact	Transparency and Communication
Construction Professionals:	Educational and Research	Construction Project Managers	Social Impact	Education and Awareness Campaigns
Suppliers and Manufacturers:	Supplier-Environmental Agencies	Builders and Contractors	Health and Safety Impact	Showcasing Success Stories
construction Project Managers:	Project Manager-Community	Homeowners and Property Developers	Technological Impact	Partnerships and Collaborations
Builders and Contractors:	Institutions-Industry	Regulatory Authorities and Inspectors	Regulatory and Legal Impact	Continuous Improvement
Environmental Agencies:	Laboratories Relationship:	Financial Institutions and Investors	Supply Chain Impact	Conflict Resolution Mechanisms

1.9 PROJECT BRIEF AND OPPORTUNITY FRAMING TEMPLATE

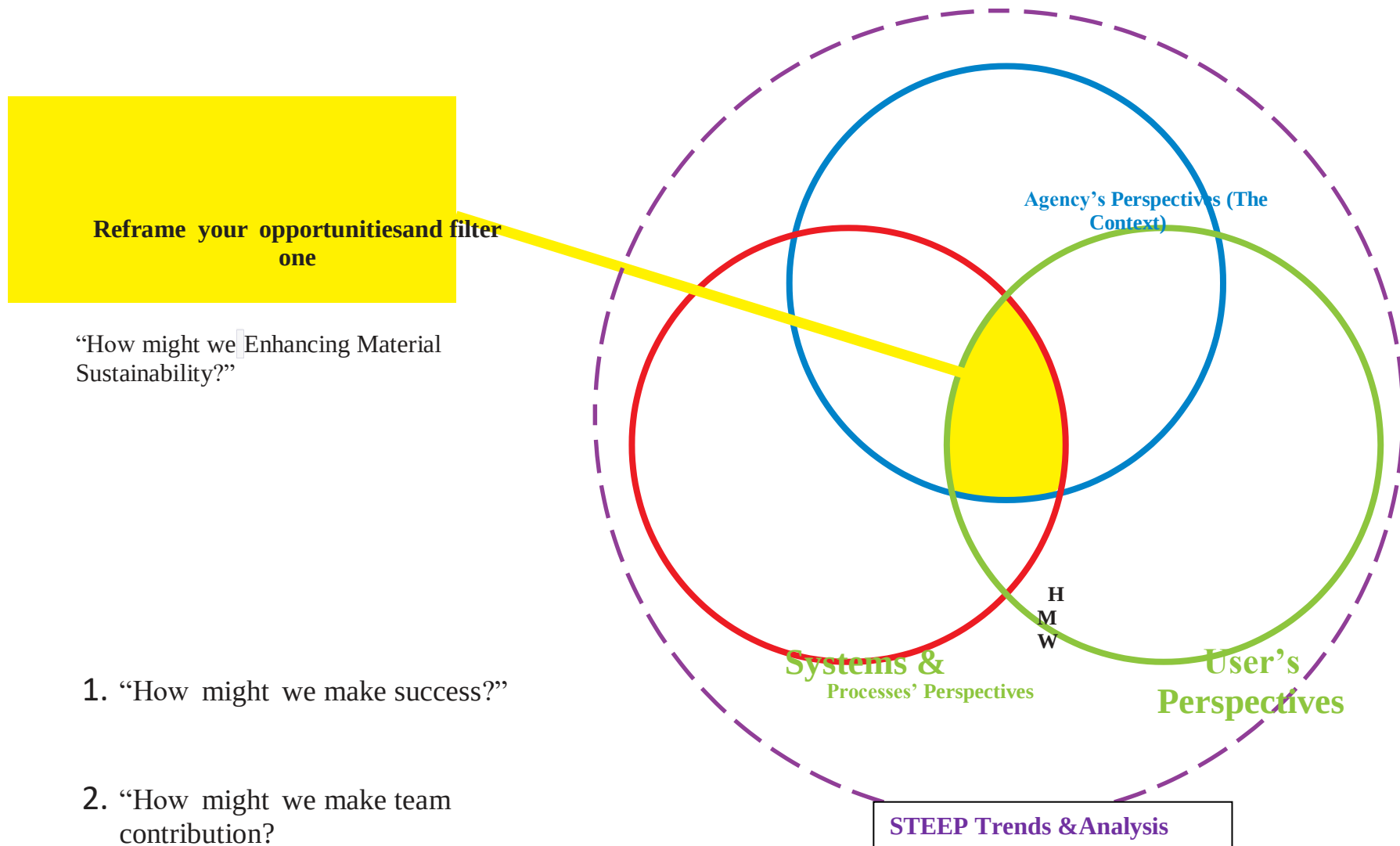
A. CHALLENGE OR PROBLEM DEFINITION		
Project Sponsor	Organization Name	SBPS
	Address and Contact	TIRUPATHI, ANDHRAPRADESH. CONTACT NO: 78905678979, 78537557565
	Contact Person(s)	t.sravan kumar, b.charan reddy, p.bharath chand, p.prashanth
Project Title	INDIAN MATERIVERSE CONSTRUCTION	
Design Challenge	Creating website for people delivering construction materials to every one who don't know about the products and price since we are providing with best price and best quality	
Design Challenge Context and Background Info	What are the issues and opportunities that inspired this design challenge many people are facing problems on material costs and quality no proper transportation for transport products lack of suggestions	
	Why does this Design Challenge matter to the organization	

A. CHALLENGE OR PROBLEM DEFINITION (contd...)	
Goals and objectives of the DesignChallenge	Promote Innovation and Adoption of Sustainable Construction Materials
Target users of this Design Challenge	Architects and Designers: Construction Professionals: Material Scientists and Researchers: Suppliers and Manufacturers: Policy Makers and Regulators:
Other key stakeholders	Government Agencies and Ministries: Industry Associations and Chambers of Commerce: Investors and Venture Capital Firms:
Previous efforts to solve this DesignChallenge	Green Building Certifications Research and Development Initiatives: Government Regulations and Incentives: Collaborative Industry Initiatives
Current activity system and process	Research institutions, universities, and material manufacturers are engaged in developing innovative sustainable materials. This includes experimenting with recycled materials, bio-based composites, and low-carbon alternatives like eco-friendly concrete.

1.10 PROJECT BRIEF AND REFRAMING PROJECT CHALLENGES TEMPLATE

B. OPPORTUNITY FRAMING	
Real issues behind this Design Challenge	Traditional construction materials contribute significantly to environmental degradation, including high carbon emissions, resource depletion, and pollution.
Inspirations from others in solving this Design Challenge	The concept of circular economy, where products and materials are reused, repurposed, and recycled, has been successfully implemented in various industries.
Teams' contributions	Teams can develop educational materials, workshops, and online platforms to raise awareness about sustainable construction practices. They can educate professionals, policymakers, and the public about the benefits of eco-friendly materials.
Success criteria	Reduction in Carbon Emissions Resource Conservation Cost-Effectiveness: Community Engagement
“HOW MIGHT WE “Opportunity/ possibility statement	HMW Reduce Carbon Emissions and Conserve Resources? HMW Make Sustainable Materials Economically Viable? HMW Engage Communities and Create Jobs?

1.11 REFRAMING THE OPPORTUNITIES TEMPLATE

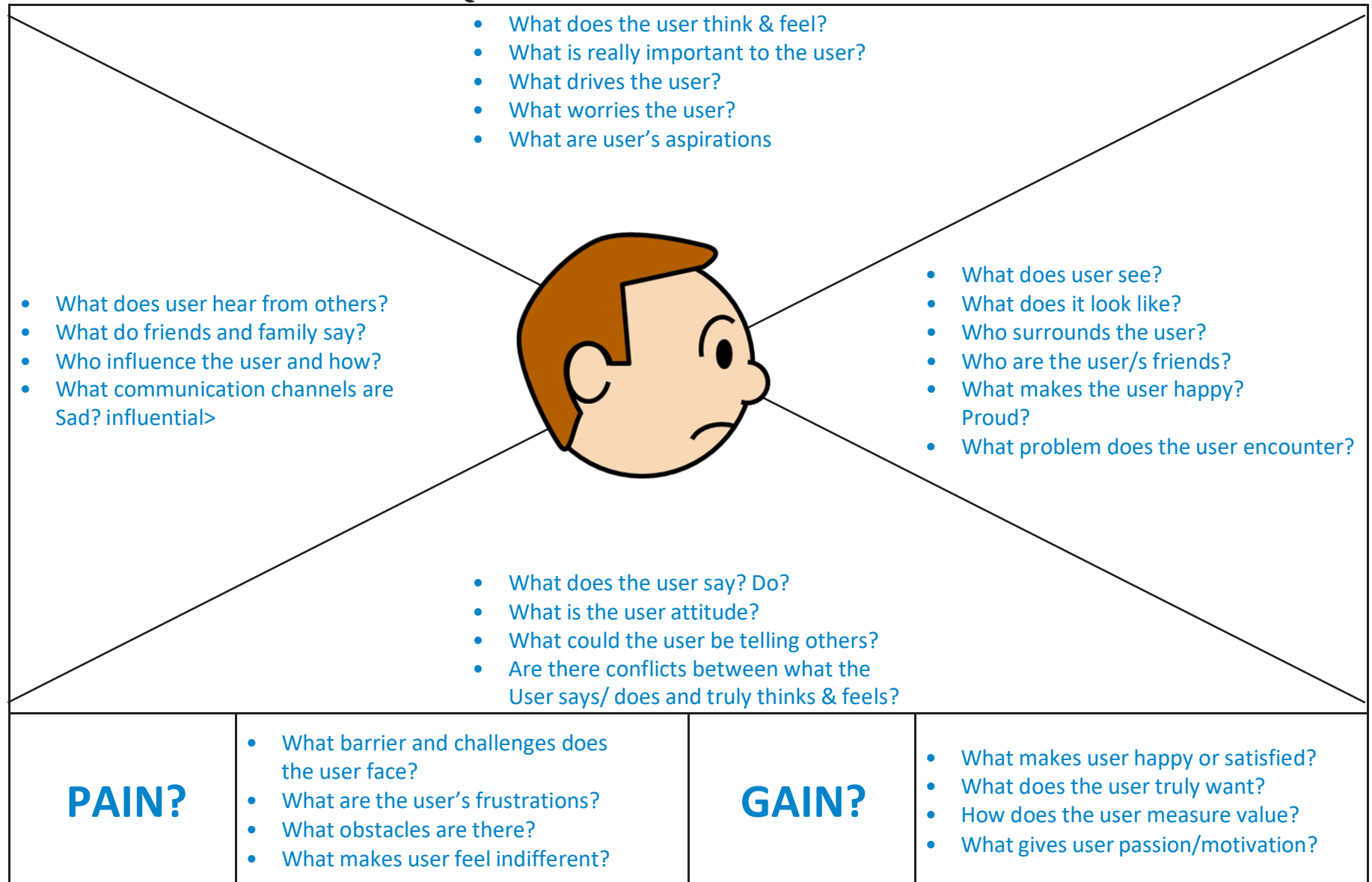


EMPATHISEPHASE

2.1 POEMS FRAMEWORK TEMPLATE

Location: chennai		Date/Time/Period: 27/3.30
Report By: imc.crp.ltd		
POEMS Framework	Field Visit and Onsite Observation	Insights
People: Who are involved? (E.g., commuters, street vendors, office workers, children, motorists, delivery men, etc.) What roles do they play? How are the people engaging with each other? How are they related? What are the relationships? What is the social context?	User Registration and Profiles: Material Listings: Location-Based Search	Sustainability is Key Environmental Impact Awareness
Objects: What artifacts are important? (E.g., bus stops signage, traffic lights, benches, etc.) What roles do they play? How are people engaging with the objects, and with their surrounding? What is interaction? How does the object influence people's behavior?	Name, description, category, price, supplier details, specifications. Add to cart, view supplier information	Sustainable Material Object Carbon Footprint Analysis Certification Object
Environment: Where is the action taking place? (e.g., public bus, road side, government office, garden, etc.) What is happening? What are the people doing? How do the people behave in this environment? How does the environment influence people's behavior? What is the mood? Ambience?	Encourage suppliers to offer eco-friendly and sustainable construction materials.	Eco-Friendly Material Sourcing Reduced Carbon Footprint
Messages & Media: What are the messages and communication media used? (e.g., signage, online materials, posters, apps, etc.) What roles do they play?	Building Tomorrow, Sustainably Today: Your Trusted Partner in Eco-Friendly Construction Materials."	Welcome Message Tagline: Environmental Focus
Services: What are the services and support systems provided? (e.g., registration, library services, hands-on guide, online booking, etc.)	Arrange for experts to conduct onsite inspections of materials to ensure they meet quality standards.	Eco-Friendly Material Sourcing Carbon Footprint Analysis

2.2 GENERATE INTERVIEW QUESTIONS



2.3 EMPATHY MAP & USER JOURNEY TO GENERATE INTERVIEW QUESTIONS

DOING	SEEING	HEARING/SAYING	FEELING/THINKING	FRUSTATION	NEEDS/WANTS
When does target User go there? How does target User go there? Sustainable materials should be cost-effective and readily available." Research new materials, attend industry events, and collaborate with suppliers. Attend community meetings, express concerns about construction projects.	Professionals gather information about sustainable materials, attend workshops, and network with experts. Innovations in lightweight concrete reduce material usage and transportation costs. Additionally, high-performance concrete formulations enhance durability, reducing the need for frequent repairs and replacements.	We want buildings that enhance our neighborhood and environment." We want buildings that enhance our neighborhood and environment."	Concerned about the local environment, want sustainable and aesthetically pleasing structures, worried about disruptions during construction.	Fear of environmental degradation, resistance to new construction projects. Attend community meetings, express concerns about construction projects, seek eco-friendly solutions for local development.	Improved local environment, pride in eco-conscious community, access to green spaces. Attend community meetings, express concerns about construction projects, seek eco-friendly solutions for local development.

2.4 USER INTERVIEW NOTES

Interviewer Name	SRAVAN KUMAR
Note Taker & Observer	BHARATH KUMAR
Interviewee Name	CHARAN REDDY
Interviewee contact details	7896578990
Date/Time of Interview	20/10/23
Interview Location	chennai
Interview's background information	
Gender	Male
Age Range	20
Profession	Na
Education Background	b.tech
Family	yes
Hobbies	Reading,playing
Likes/Dislikes	Likes humble persons
Interview Notes (Verbatim)	Observations/Quotes
Quick access to a variety of construction materials.	Implement a user-friendly interface for easy material selection.
Detailed product specifications and certifications for sustainable materials.	Provide real-time stock availability and accurate delivery estimates.
Efficient ordering process and on-time delivery to avoid project delays	Offer discounts for bulk orders to incentivize sustainable material use
Challenges:	Offer training sessions for suppliers to optimize their product listings.
Limited availability of specific eco-friendly materials.	Implement secure payment gateways and data encryption for transactions.
Concerns about material quality and authenticity.	Establish a responsive customer support system for supplier inquiries. Difficulty in comparing prices and specifications across multiple platforms.
Need for bulk ordering options for large-scale projects.	Limited awareness of new, sustainable construction materials.
	Concerns about the reliability of online reviews and ratings.

2.5 POST INTERVIEW DEFREIFPRESENTATION:

Interviewee's Goals & /or Motivation	Interviewee's Aspirations:
Efficiency: Cost-effectiveness Professional Reputation Innovation	Aspiring to be recognized as industry leaders who prioritize eco-friendly construction practices, enhancing their professional reputation.
Interviewee's Current Experience	Interviewee's Challenges &/or Pain Points
Very exciting for interviewing on the construction project on first time	Construction professionals often struggle to find a diverse range of sustainable materials, limiting their choices when planning eco-friendly projects.
3 most memorable things about the interviewee:	User Insights &/or needs
Passion for Sustainability: Innovative Solutions and Ideas: Real-Life Anecdotes and Challenges:	Users are increasingly aware of the environmental impact construction materials and seek eco-friendly alternatives. Users, especially homeowners and smaller construction projects, are often constrained by budgets.

2.6 POST INTERVIEW DISCUSSIONS: ABOUT THE PROCESS

What went well with the Interview?
The interview process generated rich and diverse user insights. Interviewers were successful in eliciting valuable information about the needs, challenges, aspirations, and motivations of stakeholders involved in the construction materials industry.
What did not go well with the Interview?
Some key stakeholders might not have participated in the interviews. Missing perspectives could result in gaps in understanding the user needs and challenges, leading to an incomplete picture of the stakeholders' requirements.
How can we do better for next Interview?

2.7 POST INTERVIEW DEFREIFPRESENTATION

END OF INTERVIEW NOTES

Summary profile of interviewee

	Insert Photo Here		Overview of interviewee (Brief demographic profile, likes /dislikes, lifestyle etc...):
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Summary profile of interviewee

Goals/Motivation: Efficiency: Streamlining the material procurement process to save time and avoid project delays. Quality: Ensuring the materials ordered are of high quality and meet project specifications.	Aspiration: To be at the forefront of innovative and sustainable construction practices, integrating cutting-edge materials into projects to reduce environmental impact and enhance building efficiency.
Current Experience: It is good to work on this project to know the middle class people problems and finding solutions for the irreplaceable question but finally we achieved it	Challenges and Pain Points: Challenge: Finding a diverse range of sustainable construction materials can be difficult, limiting options for eco-friendly projects. Pain Point: Limited availability hampers the ability to meet green construction goals and project specifications.

3 most memorable things about the interviewee: Passion for Sustainability: Innovative Solutions: Real-World Experience: Memorable Aspect	User Insight & Deep Needs: Users who prioritize eco-friendly materials are driven by a deep need for ethical fulfillment. They seek construction materials that align with their values, allowing them to contribute positively to the environment.
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2.8 DOCUMENTING INSIGHTS & NEEDS:EXAMPLE

INSIGHT #: Users who prioritize eco-friendly materials are driven by a deep need for ethical fulfillment. They seek construction materials that align with their values, allowing them to contribute positively to the environment.

Insert narrative of the user insight, needs, quotes here.

Yet, amidst their noble aspirations, users face the daunting challenge of budget constraints. This insight uncovers a fundamental human need the need for financial security. In their quest for construction materials that are eco-friendly and economical, users are, in essence, searching for stability. They need materials that not only fit snugly within their financial boundaries but also offer the promise of longevity, ensuring the financial security of their projects.

Need: Stakeholders need access to a wide range of sustainable construction materials. These materials should be eco-friendly, energy-efficient, and sourced responsibly. Access to such materials fulfills the need for environmentally conscious construction practices.

Quotes 1: "Sustainability is not just a trend; it's a responsibility. Choosing eco-friendly construction materials is our way of giving back to the planet."

Quotes 2: "Quality construction begins with quality materials. It's not just about building structures; it's about building trust and reliability."

PERSONA CANVAS	Persona Name: SOHAN	
Demographic Profile: Age:43 Gender:male Home: yes Family:joint family	Goals: Showcasing their range of sustainable construction materials to a wider audience.	Deep Need Statement: I need construction materials that align with my ethical values, allowing me to contribute positively to the environment and build projects that reflect my commitment to sustainability."
Hobbies/Likes/Dislikes:reading book/loyalty ,honest charcter	Motivation/Aspiration: To be leaders in the market, providing a diverse range of sustainable construction materials that are widely adopted by professionals and end-users alike. Challenge: Integrating new sustainable materials with existing structures can be complicated. Pain Point: Compatibility issues can result in construction delays and additional costs for modifications.	Diversity of Needs: Contractors, architects, and homeowners require access to a wide variety of construction materials ranging from basic commodities like cement and steel to specialized materials like eco-friendly alternatives and innovative smart materials. 
Social & Family Lifestyle: Homeowners and families are influenced by the desire for a comfortable and aesthetically pleasing living environment. They often prioritize materials and designs that enhance the quality of family life, such as durable and easy-to-maintain materials suitable for children and pets.	Behavior: Stakeholders extensively research construction materials online, comparing prices, quality, and eco-friendly features before making a decision.	

EXPERIMENTPHASE

3.1 SCAMPER WORKSHEET

S	Substitute	(What can you substitute? What or who could you use instead?) Substitute traditional concrete with engineered wood products like cross-laminated timber for certain construction applications, reducing the carbon footprint.
C	Combine	(What can you combine or bring together?) Combine recycled steel, reclaimed wood, and eco-friendly concrete to create a composite material that is both durable and environmentally friendly, suitable for various construction applications.
A	Adapt	(What can you adapt for use as a solution?) Modify existing materials to make them more sustainable and environmentally friendly. Example: Adapt traditional concrete formulations by incorporating industrial by-products like fly ash or slag, reducing the carbon footprint while maintaining structural integrity.
M	Modify	(Can you modify or change the item in some ways?) Alter the composition of existing materials to enhance their sustainability and performance. Example: Modify traditional insulation materials by incorporating recycled fibers or aerogels, improving thermal efficiency and reducing the ecological impact.
P	Put to other uses	(What other or new ways could this be used?) New Applications for Materials: Use: Explore new applications for sustainable materials in construction and related industries.
E	Eliminate	(What can you get rid of?) Remove asbestos-containing materials completely from construction projects, replacing them with safer and more environmentally friendly alternatives.
R	Reverse	(What could you rearrange or change around?) Rearrange the combination of bamboo, recycled steel, and low-carbon concrete in various ways to create innovative composite materials with unique properties suitable for different construction applications.

3.2 RECONNECTING WITH OUR PERSONAS

Behaviors	Aspirations	Motivations	Challenges	Pain Points
Construction Professionals (Contractors, Architects, Builders) Material Suppliers and Manufacturers: Environmental Organizations and NGOs:	Industry Leadership in Sustainability: Empowering Informed Decision-Making: Driving Innovation in Eco-Friendly Materials:	Enhancing Reputation and Client Satisfaction Market Differentiation and Increased Demand environmental Conservation and Climate Mitigation Environmental Regulations and Public Welfare	Limited Awareness and Education: Higher Initial Costs: Supply Chain Complexity: Certification and Standards Variability:	Limited Availability and Variety: Higher Initial Costs: Complex Certification and Standards: Resistance to Change:

User insights:

Users, especially construction professionals and buyers, are highly sensitive to costs. Budget constraints often influence their choice of construction materials.

Deep user needs:

Users have a fundamental desire to contribute positively to the environment and reduce their ecological impact.

Constraints:

Sustainable materials often have higher upfront costs, making them financially challenging for budget-conscious construction projects.

ENGAGEPHASE

4.1 STORY BOARD CANVAS

Directions:

- Review your persona, the ideal user experience journey and the prototype you have created.
- Brainstorm how you intend to illustrate your persona's story in a coherent manner including the narratives.
- Use the story board template / canvas provided and start drawing.

TITLE: INDIAN MATERIVERSE CONSTRUCTION

Here we are going to derieve the problem facing by people about their home construction and we find the problem and derive the goal and aim and after we gone up some interragation andinterviews and we sort out and found a solution by creating website for buying all the construction materials and field materials in this website we are providing each material what we need for the building home including bricks, cement, iron, steel, wood, hardware tools, electrical tools, everything what we need by this website we can buy with very low cost and good product quality.

4.2 STORY BOARDING CANVAS

BEGINNING MIDDLE END				
The Persona	The setting	The Problems	The Solutions	The Resolutions
Personalization in the context of a construction materials delivery platform involves tailoring the user experience based on individual preferences, behavior, and project requirements. Implementing personalization can significantly enhance user engagement, satisfaction, and the likelihood of promoting sustainable construction materials.	The setting for implementing the strategies and personalization features for promoting sustainable construction materials delivery would be an online platform specifically designed for construction professionals, contractors, architects, builders, and other stakeholders involved in construction projects.	Not all suppliers may offer a wide range of sustainable construction materials. Limited supplier adoption can restrict the variety of eco-friendly products available on the platform. Sustainable materials, especially those with certifications, can be more expensive than traditional alternatives. Budget constraints might deter construction professionals from choosing eco-friendly options.	Actively engage with suppliers, provide incentives for adopting sustainable practices, and offer support for obtaining environmental certifications. Encourage the development of eco-friendly product lines. Educate users about the long-term cost benefits of sustainable materials, including energy efficiency and reduced maintenance costs. Negotiate bulk discounts with suppliers to make eco-friendly options more affordable.	Encourage suppliers to adopt sustainable practices by offering them financial incentives, marketing support, and exposure on the platform for eco-friendly products. Establish partnerships with suppliers committed to sustainable sourcing and production. Cost Considerations: Lack of Awareness:

EVOLVEPHASE

5.1 STRATEGIC REQUIREMENT TEMPLATE

Strategic Requirements	The Big Idea or the Idea Concept (Main Solution to be delivered)
Key Solution Components of the Big Idea	Implement a comprehensive digital platform that integrates suppliers, manufacturers, logistics providers, and construction sites.
Capabilities Required to Deliver this Solution Component	In-depth knowledge of supply chain processes, including procurement, inventory management, logistics, and supplier collaboration, to optimize the entire materials delivery system.
Current Organizational Assets & capabilities to be Leveraged	Negotiate favorable terms, secure timely deliveries, and collaborate for bulk discounts, ensuring a steady supply of materials.
Development Strategy to develop this capability (if needed)	Skills Assessment and Training: Strategic Partnerships and Collaborations: Prototyping and Agile Development: User-Centric Design
Requirements and cost to develop (high / med / low)	Its requires high level application to build and it costa upto 50 millions
External Sources of Expertise (potential partnership)	Web Development Agencies: E-commerce Solution Providers: Digital Marketing Agencies: Logistics and Supply Chain Consultants:

5.2 EVOLVING THE PROCESS FOR DELIVERY

Key Solution Component	Login website using your credentials and check for products and select and order the product by paying money					
(Input Your Solution Component 1)	Open website	Search what type of product you want	Select the product you want	Go to cart and click on place order	Place order	Track the order
	Enter your login details	Click on the submemu	Add to cart or whishlist	Enter details and address for shipping	Your shipment is processing	Your order is on the way

5.3 IMPACT EVALUATION INDICATORS

CRITERIA	INDICATORS & MEASUREMENT	STAKEHOLDERS(CONSTRUCTORS)
Social Value Creation	1.... Implementation 2....Engage with local communities where your materials are sourced or delivered. Contribute to community projects, support local schools, or sponsor events.	1.... Customers: 2....Suppliers and Manufacturers: Timely payments, fair business practices, consistent orders, and adherence to quality standards.
Stakeholder Satisfaction	1.... Strategy: Provide high-quality construction materials, ensure on-time and accurate deliveries, offer excellent customer service,	1....Tactics: Implement user-friendly website design, optimize the checkout process, offer competitive pricing
Solution Sustainability	1.... Source and promote the use of eco-friendly and sustainable construction materials. This includes recycled materials, renewable resources, and products with low environmental impact.	1.... Invest in research and development to discover new eco-friendly materials and innovative sustainable practices. Regularly assess the environmental impact of operations.
Solution Scalability	1.... Ensure website performance and responsiveness during high traffic periods. Scale server resources up or down based on demand.	1.... Utilize scalable cloud hosting services that allow you to increase server resources as needed. Implement Content Delivery Networks (CDNs) to distribute content efficiently.

5.4 ACTION PLANNING TO ADVANCE THE DESIGN CHALLENGE PROJECT

Idea	Objectives	Responsibility	Implementation	Resources	Completion
What idea for implementation	Why is this idea important? Values and benefits	Who will lead this?	How will this be Implemented?	What capability and resources are needed?	When will this be completed?
Create immersive virtual reality showrooms that allow customers to explore construction materials in a digital environment. This innovative approach leverages VR technology to enhance customer experiences, provide valuable product information, and facilitate purchasing decisions.	Enhanced Customer Engagement: VR showrooms provide an immersive and memorable shopping experience, increasing customer engagement and satisfaction. Better Informed Purchases: Customers can thoroughly examine products, understand their features, and visualize them in their spaces	host virtual events or webinars within the VR showroom, showcasing new products, industry trends, and expert talks. Allow attendees to interact with presenters, ask questions, and explore products in real-time.	Develop VR Content: Create detailed 3D models and high-resolution textures of construction materials, including various types of flooring, tiles, paints, and other products. Use VR development tools and software to design interactive and visually appealing virtual showrooms.	Technical Expertise: Content Creation: VR Hardware: E-commerce Integration: Virtual Assistant Integration: User Experience and Testing: Marketing and Promotion: Data Analysis: Legal and Compliance:	Planning and Conceptualization (1-2 months): Development and Content Creation (3-6 months): Testing and Iteration (1-2 months): Launch and Marketing (1-2 months):

5.5 IDENTIFYING QUICK WIN

What is this Quick Win (1) about?	Simplify the order placement process by reducing the number of steps and form fields. Implement an intuitive and user-friendly interface for quick material selection and checkout.
What are the success indicator(s)? How would it (these) be measured?	The percentage of website visitors who take the desired action (e.g., placing an order, signing up for a newsletter).
What are the resources / staff trainings needed?	Training in digital marketing strategies, SEO techniques, content marketing, social media management, and online advertising platforms (e.g., Google Ads, Facebook Ads).
Who will lead this Quick Win implementation?	Simplify the order placement process by reducing the number of steps and form fields. Implement an intuitive and user-friendly interface for quick material selection and checkout.
What are the key steps needed to implement this Quick Win? What is the timeline till completion?	Assess the current state of the website, identifying strengths, weaknesses, opportunities, and threats (SWOT analysis).
When will be the status or progress update?	Weekly Updates: Bi-Weekly Updates: Monthly Reviews: Milestone Updates:
When will this be completed?	When the above all requirements all completed successfully
How would the Success be communicated?	interview customers, gather testimonials, and compile data to create compelling narratives. Share these stories on the website, social media, and in marketing materials.

5.6 STORY BOARDING CANVAS

Design Challenge	How Might We...	Solution Concept	We design a website for delivering construction materials with transportation
Persona: Alex works as a sustainability officer in an architectural firm that focuses on eco-friendly and sustainable building designs. Alex is responsible for sourcing environmentally responsible construction materials for the firm's projects.			Value Propositions to Organizations/Agency: Project-specific Recommendations: Tailored recommendations based on project requirements, ensuring optimal material choices for each project phase.
Deep Needs: Users want confidence that the materials they choose are of high quality, authentic, and suitable for their specific projects.	Value Propositions to Target Users: Efficient sourcing and bulk discounts, real time inventory, reliable deliveries, Curated Materials.		Budget-Friendly Options: Access to a diverse range of materials, including economical alternatives without compromising quality.
Gains: Lower material costs due to bulk discounts, competitive pricing, and budget-friendly alternatives, maximizing the value of construction budgets.	User Need(Problem) Solvers: Extensive Catalog, Quality Assurance, One-Stop Solution, Quick Ordering, Competitive Pricing, Volume Discounts:		
Pains: Limited Availability and Variety: Higher Initial Costs: Complex Certification and Standards: Resistance to Change:	Gain Creators: User-Friendly Interface, Real-Time Inventory Updates	Pain Relievers: Simplified Ordering: Intuitive and streamlined online platform for quick and effortless bulk material orders.	

5.7 M–A–R–S FRAMEWORK

Use the MARS framework to understand the people's behavior in the face of the change and innovation Motivation It is about the Why / the Will to change				
		Ability	Role	Systems
		It is about the Tools and Skills	It is about defining Roles/To Rs	It is about the Support system (Resources)
Engage team in conversation to connect and to empathize. Listen to welcome the truth and to gain trust	Think	Sustainable Material Metrics Environmental Impact Assessment Sustainable Sourcing	Responsibilities Strategic Planning Supplier Vetting	Expert Consultation Comprehensive Database Research and Development Team
	Feel	Optimism Responsibility Inspiration	Inspiration Determination Responsibility	Anticipation and Readiness Determination and Commitment
	Do	Passion and Purpose Hopeful Anticipation	Passion and Purpose Optimism and Hope	Strategic Vision Resource Alignment Resource Synergy

5.8 WHAT IS OUR CHANGE MANAGEMENT PLAN?

Vision (Reasons)for change	Staff Engagement	Communicate visionfor change	Implementation Plan	Empower people for change	Create Quick Wins
Efficiency and Convenience: Transparent Pricing and Options: Reliable and Timely Deliveries: Sustainability and Green Practices: Digital Transformation and Innovation: Community Engagement and Support: Safety and Compliance:	Supply Chain and Logistics Team: Website Development Team: Project Manager: Technology and Innovation Officer: Compliance and Quality Assurance Team: Marketing and Sales Team: Customer Service and Support Team:	Efficiency and Convenience: We aim to create a user-friendly platform, streamlining the entire procurement process. Your time is valuable, and our platform will empower you to find, order, and track construction materials with unmatched ease. Reliability and Innovation: Timely deliveries and cutting-edge technology will be our hallmark. Real-time tracking, smart recommendations, and continuous innovation will redefine what it means to deliver exceptional customer service.	Market Research and Supplier Partnerships: Conduct in-depth market research to identify customer needs and preferences. Forge partnerships with reliable suppliers and negotiate contracts for a diverse range of high-quality materials, including eco-friendly options.	Clearly communicate the vision, objectives, and expected outcomes of the construction material delivery website. Ensure everyone understands the purpose and benefits of the initiative. Create an innovation recognition program where employees are rewarded for suggesting and implementing innovative ideas.	Launch a basic version of the website with essential features like easy navigation, product listings, and a simplified checkout process. Integrate a live chat support system on the website for instant customer assistance. Introduce limited-time promotions or discounts for the first few weeks after the website launch.

Change management in the context of promoting sustainable construction materials delivery involves a systematic approach to drive organizational and industry-wide transitions toward eco-friendly practices. Here's a structured change management plan for the construction materials delivery platform:

1. Leadership Alignment:

- **Define Clear Vision:** Establish a clear vision of becoming a leader in sustainable construction materials delivery.
- **Leadership Buy-In:** Gain commitment from top management, emphasizing the importance of sustainability in the platform's long-term strategy.

2. Stakeholder Engagement:

- **Stakeholder Workshops:** Conduct workshops involving key stakeholders to understand their concerns, expectations, and suggestions regarding sustainable practices.
- **Feedback Mechanisms:** Implement continuous feedback mechanisms to involve stakeholders in decision-making processes and demonstrate that their opinions are valued.

3. Education and Training:

- **Internal Training:** Educate internal teams about sustainable materials, certification processes, and industry best practices.
- **User Training:** Provide educational resources to platform users, promoting the benefits of sustainable materials and explaining how to identify and choose eco-friendly options.

4. Policy and Process Alignment:

- **Develop Sustainable Guidelines:** Establish clear guidelines and standards for listing materials on the platform, focusing on sustainability criteria and certifications.
- **Policy Integration:** Integrate sustainability policies into procurement, supplier onboarding, and quality control processes, ensuring adherence to eco-friendly standards.

5. Technological Integration:

- **Platform Features:** Integrate features highlighting sustainable materials, such as filters, badges, and detailed product information related to environmental impact.
- **Data Analytics:** Utilize data analytics to monitor user behavior, identifying trends and preferences related to sustainable materials. Use this data to enhance offerings and user experience.